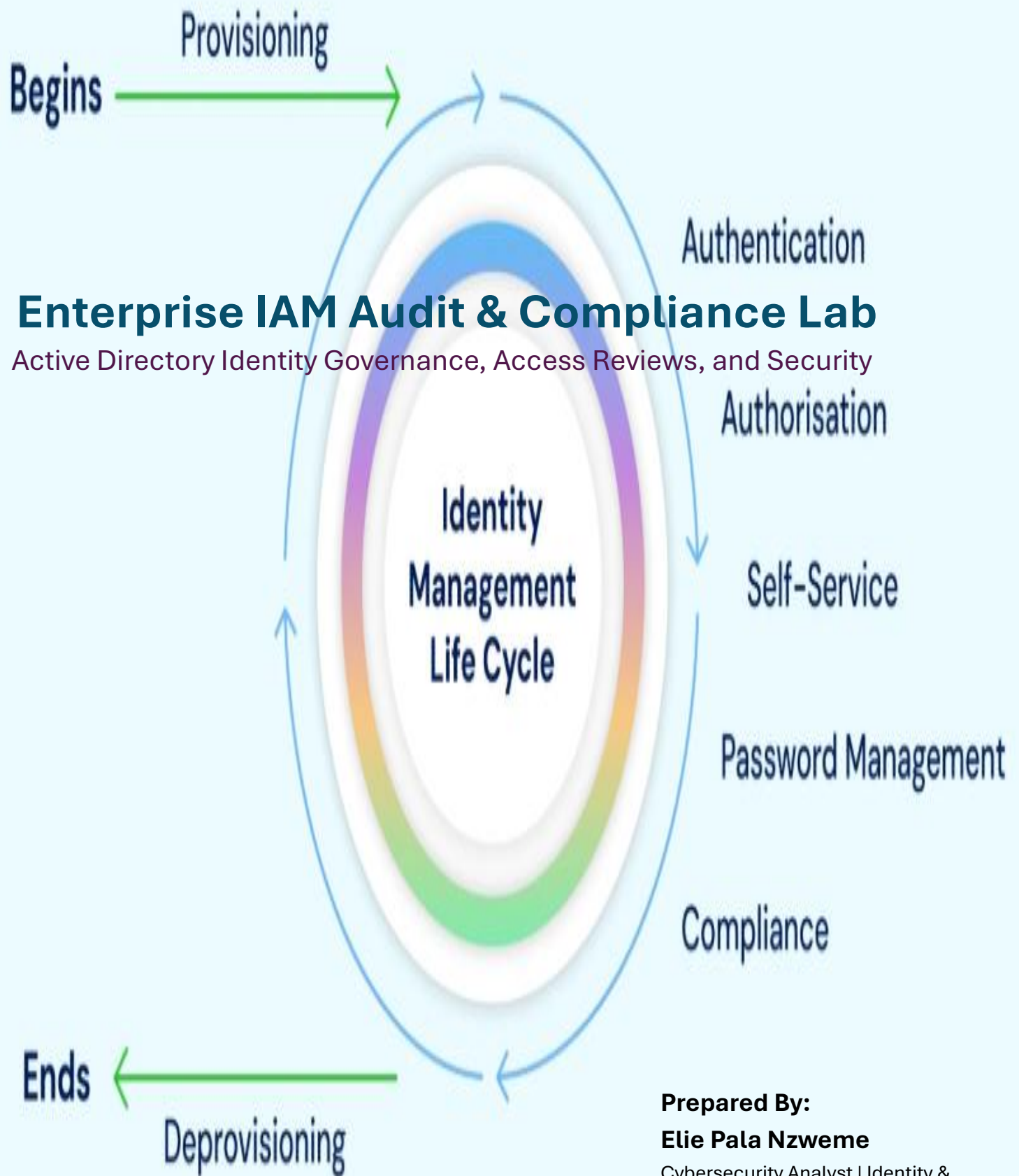




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Enterprise IAM Audit & Compliance Lab

1. Overview

The Enterprise IAM Audit & Compliance Lab demonstrates Identity and Access Management (IAM) auditing and reporting using Microsoft Active Directory and PowerShell automation.

The project simulates common IAM and compliance review activities performed by Security Operations, IAM Analysts, System Administrators, and Governance, Risk, and Compliance (GRC) teams. Multiple audit scripts were developed to identify inactive accounts, disabled accounts, locked accounts, privileged users, and password management risks.

Each script generates CSV reports that can be used for security reviews, compliance assessments, and access governance activities.

2. Project Objectives

- Audit Active Directory user accounts
- Identify inactive user accounts
- Identify disabled accounts
- Detect locked-out accounts
- Review privileged access assignments
- Audit password lifecycle information
- Automate reporting with PowerShell
- Generate compliance-ready CSV reports

3. Technologies Used

Identity & Access Management

- Windows Server 2025
- Active Directory Domain Services (AD DS)
- PowerShell
- Group Policy
- Windows 11

- CSV Reporting
- RSAT (Remote Server Administration Tools)

Operating Systems

- Windows Server 2025
- Windows 11

Scripting & Automation

- PowerShell
- Active Directory PowerShell Module

Reporting

- CSV Reporting
- Microsoft Excel

4. Lab Environment

Domain Controller

System Role

DC01 Active Directory Domain Controller

Client System

System	Role
Client01	PowerShell Audit Workstation

Client01: Install RSAT AD Tools

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-WindowsCapability -Name RSAT* -Online | Select-String ActiveDirectory
PS C:\Users\Administrator> Add-WindowsCapability -Online -Name Rsat.ActiveDirectory.DS-LDS.Tools~~~~~0.0.1.0

Path          :
Online        : True
RestartNeeded : False
```

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-WindowsCapability -Online | Where-Object {$_.Name -like "*ActiveDirectory*"}

Name : Rsat.ActiveDirectory.DS-LDS.Tools~~~~~0.0.1.0
State : Installed
```

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> hostname
Client01
PS C:\Users\Administrator> whoami
corp\administrator
PS C:\Users\Administrator> ping DC01

Pinging DC01.Corp.local [192.168.10.10] with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\Administrator> |
```

Verify AD Module

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADDomain

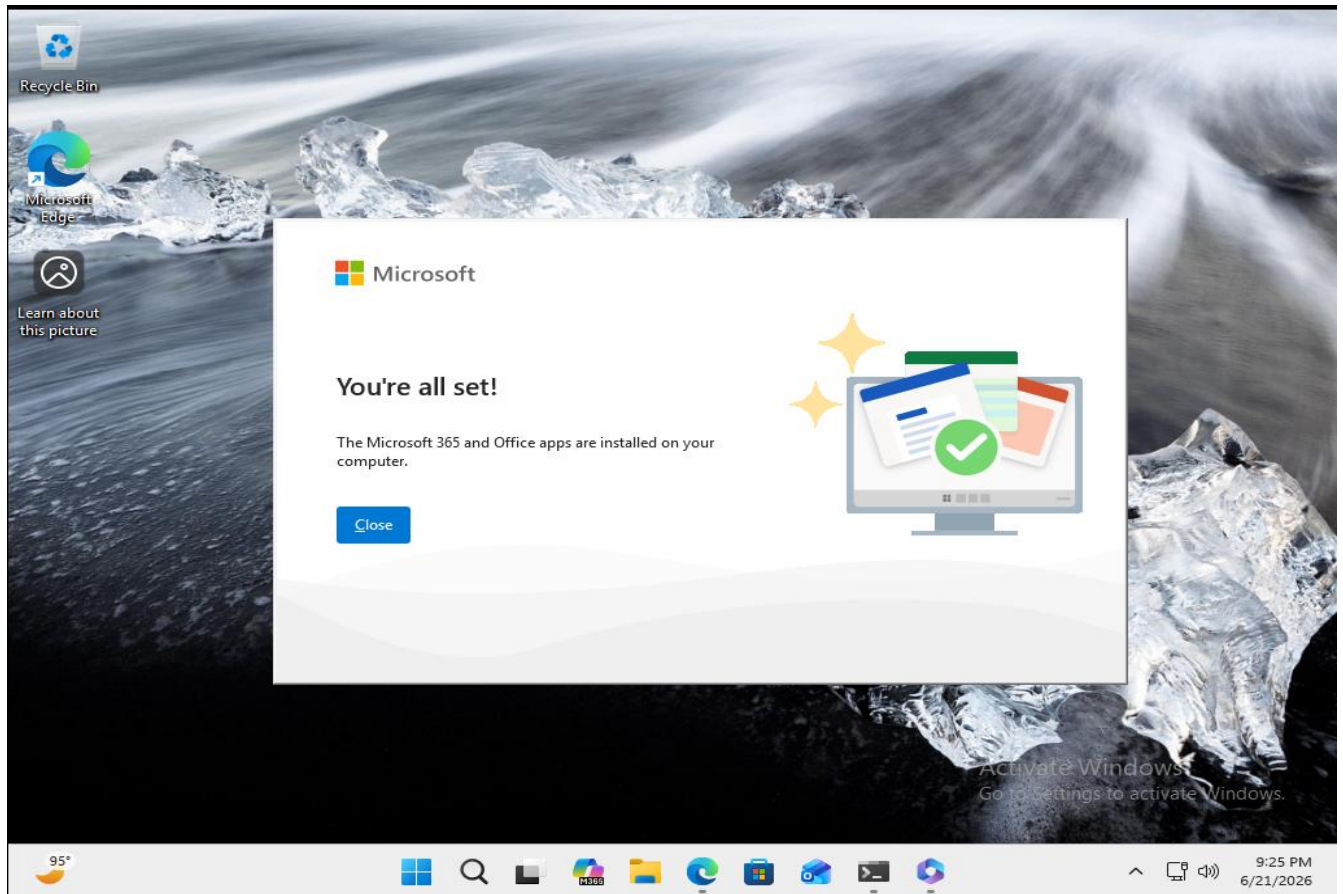
AllowedDNSSuffixes      : {}
ChildDomains            : {}
ComputersContainer      : CN=Computers,DC=Corp,DC=local
DeletedObjectsContainer : CN=Deleted Objects,DC=Corp,DC=local
DistinguishedName       : DC=Corp,DC=local
DNSRoot                : Corp.local
DomainControllersContainer : OU=Domain Controllers,DC=Corp,DC=local
DomainMode              : Windows2025Domain
DomainSID               : S-1-5-21-3377002229-886130070-2046376786
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=Corp,DC=local
Forest                 : Corp.local
InfrastructureMaster    : DC01.Corp.local
LastLogonReplicationInterval :
LinkedGroupPolicyObjects : {CN={31B2F340-016D-11D2-945F-00C04FB984F9},CN=Policies,CN=System,DC=Corp,DC=local}
LostAndFoundContainer   : CN=LostAndFound,DC=Corp,DC=local
ManagedBy              :
Name                   : Corp
NetBIOSName            : CORP
ObjectClass             : domainDNS
ObjectGUID              : 27b81a9b-1bc3-4734-bb99-070801f81079
ParentDomain            :
PDCEmulator            : DC01.Corp.local
PublicKeyRequiredPasswordRolling : True
QuotasContainer         : CN=NTDS Quotas,DC=Corp,DC=local
ReadOnlyReplicaDirectoryServers : {}
ReplicaDirectoryServers : {DC01.Corp.local}
RIDMaster              : DC01.Corp.local
SubordinateReferences   : {DC=ForestDnsZones,DC=Corp,DC=local,DC=DomainDnsZones,DC=Corp,DC=local,CN=Configuration,DC=Corp,DC=local}
SystemsContainer       : CN=System,DC=Corp,DC=local
UsersContainer         : CN=Users,DC=Corp,DC=local
```

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Import-Module ActiveDirectory
PS C:\Users\Administrator> Get-Module -ListAvailable ActiveDirectory

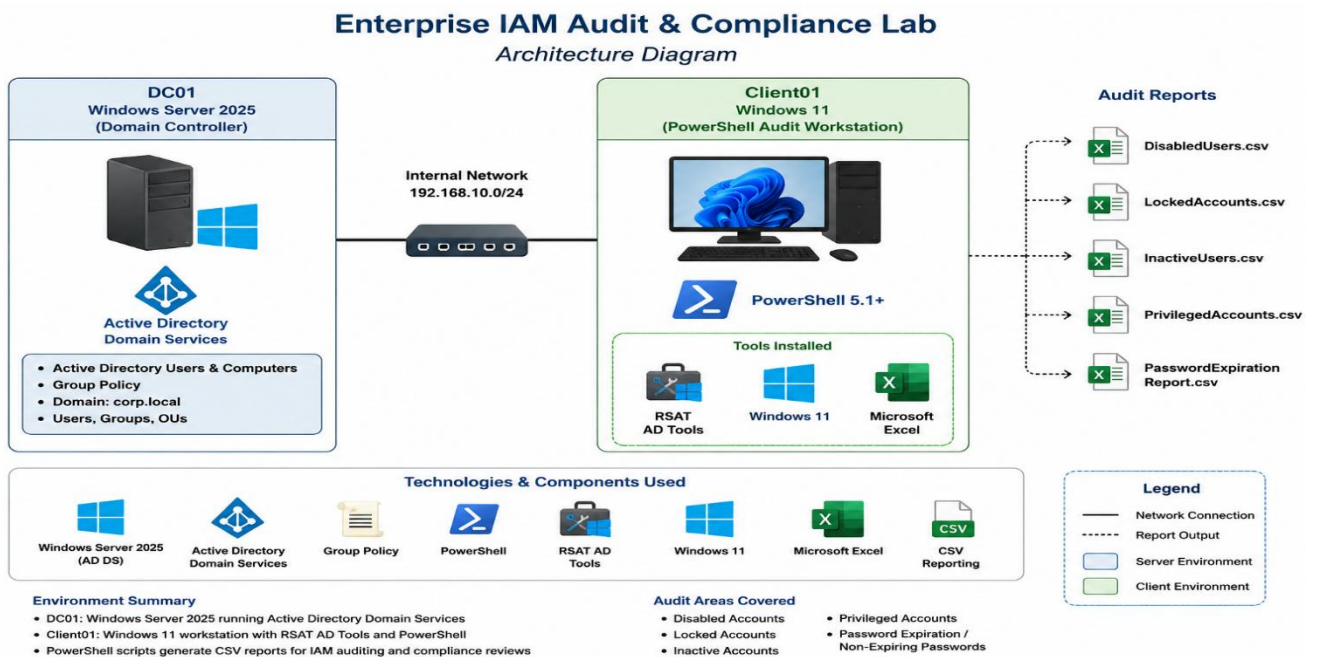
Directory: C:\WINDOWS\system32\WindowsPowerShell\v1.0\Modules

ModuleType Version Name ExportedCommands
-----
Manifest 1.0.1.0 ActiveDirectory {Add-ADCentralAccessPolicyMember, Add-ADComputer...
```

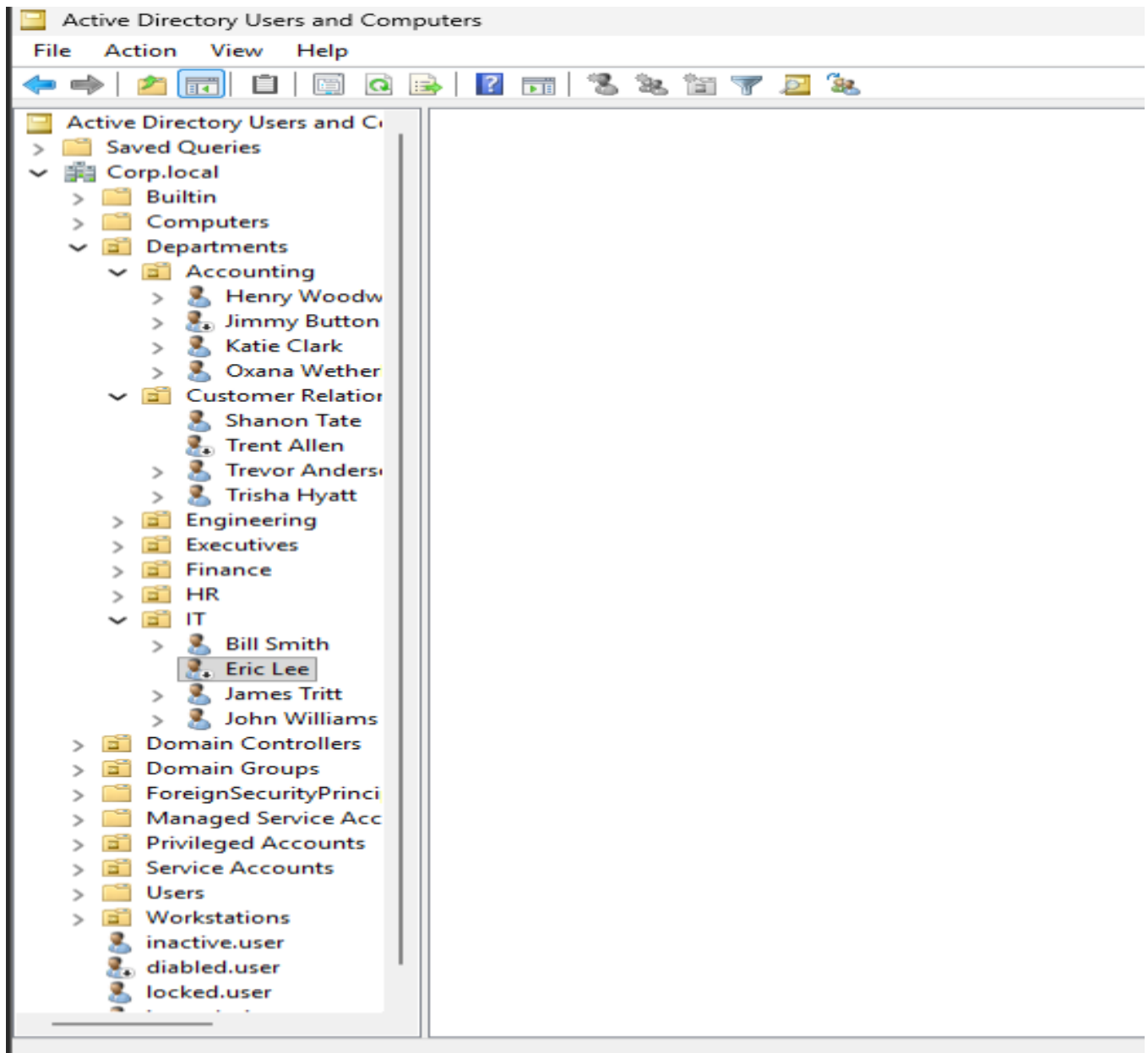
Microsoft Office Installed in Client01:



5. Architecture Diagram



Users in DC01:



6. Audit Scripts

DisabledUsers.ps1



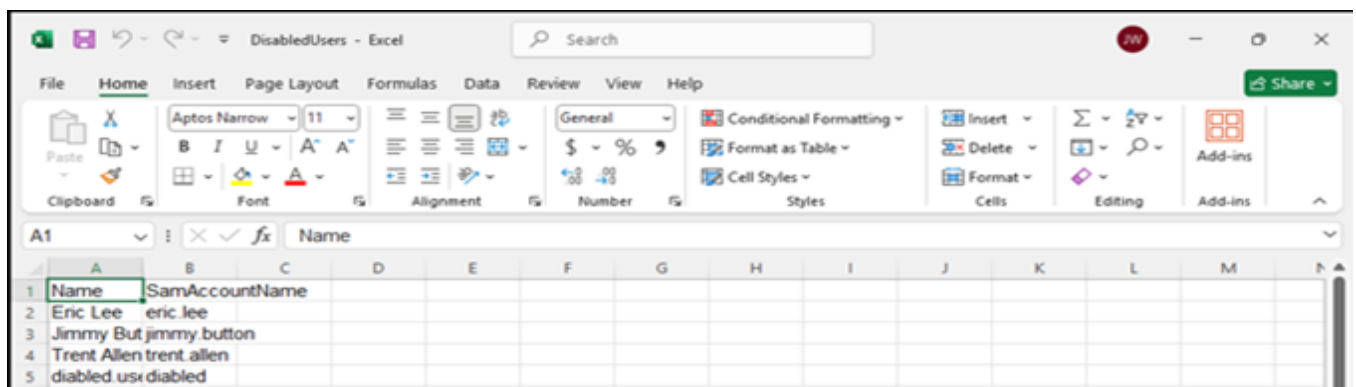

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> cd C:\IAM-Audit-Lab\Scripts
PS C:\IAM-Audit-Lab\Scripts> .\DisabledUsers.ps1
PS C:\IAM-Audit-Lab\Scripts> Import-Csv C:\IAM-Audit-Lab\Reports\DisabledUsers.csv

Name           SamAccountName
-----
Eric Lee       eric.lee
Jimmy Button   jimmy.button
Trent Allen    trent.allen
disabled.user   disabled
```

Identifies disabled Active Directory user accounts and exports results to a CSV report.

Output

DisabledUsers.csv



	Name	SamAccountName
1	Eric Lee	eric.lee
2	Jimmy Button	jimmy.button
3	Trent Allen	trent.allen
4	disabled.user	disabled

LockedAccounts.ps1

Identifies currently locked Active Directory accounts.

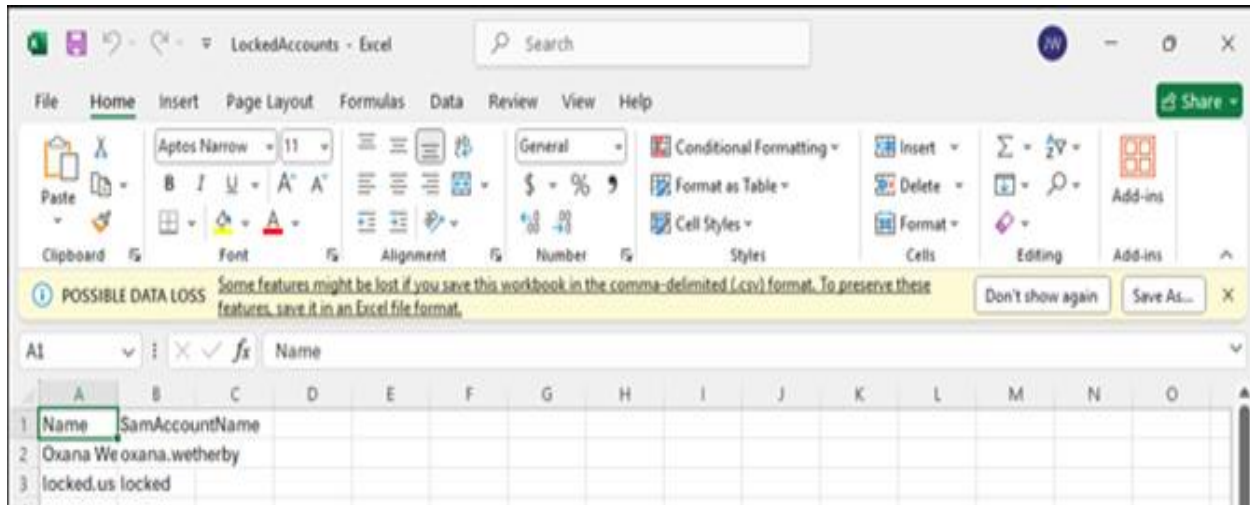
```
DisabledUsers.ps1  LockedAccounts.ps1  InactiveUsers.ps1
File Edit View
Import-Module ActiveDirectory
Search-ADAccount -LockedOut -UsersOnly |
Select Name, SamAccountName |
Export-Csv C:\IAM-Audit-Lab\Reports\LockedAccounts.csv -NoTypeInformation
```

```
Administrator: Windows PowerShell
PS C:\IAM-Audit-Lab\Scripts> cd C:\IAM-Audit-Lab\Scripts
PS C:\IAM-Audit-Lab\Scripts> .\LockedAccounts.ps1
PS C:\IAM-Audit-Lab\Scripts> Import-Csv C:\IAM-Audit-Lab\Reports\LockedAccounts.csv

Name           SamAccountName
-----
Oxana Wetherby  oxana.wetherby
locked.user     locked
```

Output

LockedAccounts.csv



InactiveUsers.ps1

Identifies inactive user accounts based on logon activity and configurable inactivity thresholds.

```
File Edit View

Import-Module ActiveDirectory

Get-ADUser -Filter * -Properties LastLogonDate |
Where-Object {
    $_.Enabled -eq $true -and
    $_.SamAccountName -notin @(
        "Guest",
        "krbtgt",
        "svc_backup",
        "svc_splunk",
        "svc_sql",
        "ITAdmin",
        "iam"
    ) -and
    (
        $_.LastLogonDate -lt (Get-Date).AddDays(-90) -or
        $_.LastLogonDate -eq $null
    )
} |
Select Name, SamAccountName, LastLogonDate |
Export-Csv C:\IAM-Audit-Lab\Reports\InactiveUsers.csv -NoTypeInformation

Ln 21, Col 73 | 524 characters | Plain text | 100% | Windows (CRLF) | UTF-8
```

```
Administrator: Windows Powe + -
```

```
PS C:\IAM-Audit-Lab\Scripts> .\InactiveUsers.ps1
PS C:\IAM-Audit-Lab\Scripts> Import-Csv C:\IAM-Audit-Lab\Reports\InactiveUsers.csv
```

Name	SamAccountName	LastLogonDate
Bill Smith	bill.smith	
James Tritt	james.tritt	
Janice Smith	janice.smith	
Jennifer Sorensen	jennifer.sorensen	
Julie Lloyd	julie.lloyd	
Katie Clark	katie.clark	
Henry Woodward	henry.woodward	
Kevin Jones	kevin.jones	
Lee McCrae	lee.mccrae	
Happy Sunday	happy.sunday	
Heather Summer	heather.summer	
Alyssa Jensen	alyssa.jensen	
Amy Wilcox	amy.wilcox	
Anthony Edwards	anthony.edwards	
Andrea Forsythe	andrea.forsythe	
Benjamin Washington	benjamin.washington	
Kathy Arnold	kathy.arnold	
Ted Williams	ted.williams	
William Trent	william.trent	
Shanon Tate	shanon.tate	
Trevor Anderson	trevor.anderson	
Trisha Hyatt	trisha.hyatt	
Test user	testuser	
inactive.user	inactive	
locked.user	locked	

PrivilegedAccounts.ps1

Identifies members of privileged Active Directory groups such as Domain Admins.

```
File Edit View

Import-Module ActiveDirectory

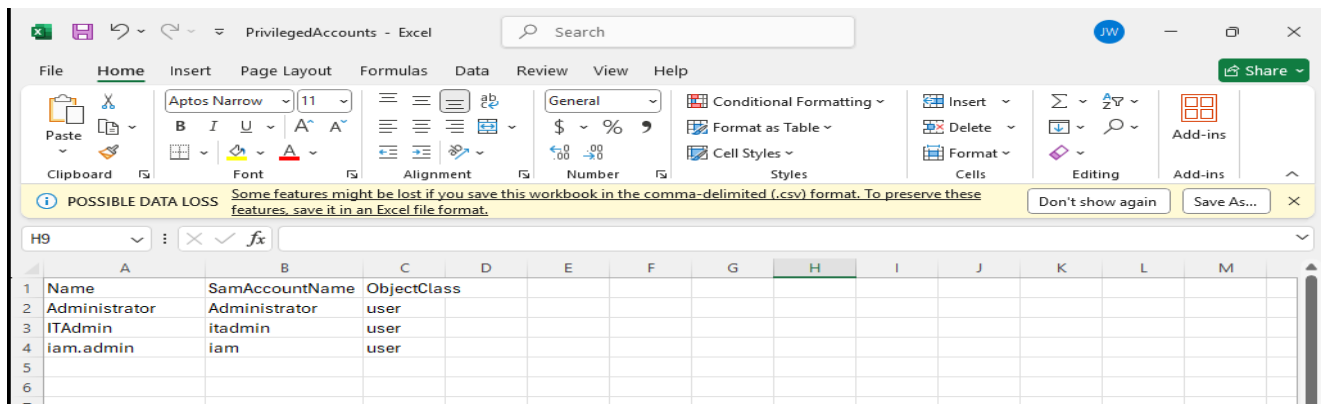
Get-ADGroupMember "Domain Admins" |
Select Name, SamAccountName, ObjectClass |
Export-Csv C:\IAM-Audit-Lab\Reports\PrivilegedAccounts.csv -NoTypeInformation
```

```
Administrator: Windows PowerShell
PS C:\IAM-Audit-Lab\Scripts> .\PrivilegedAccounts.ps1
PS C:\IAM-Audit-Lab\Scripts> Import-Csv C:\IAM-Audit-Lab\Reports\PrivilegedAccounts.csv
```

Name	SamAccountName	ObjectClass
Administrator	Administrator	user
ITAdmin	itadmin	user
iam.admin	iam	user

Output

PrivilegedAccounts.csv



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Name	SamAccountName	ObjectClass										
2	Administrator	Administrator	user										
3	ITAdmin	itadmin	user										
4	iam.admin	iam	user										
5													
6													
7													

PasswordExpirationReport.ps1

Audits password lifecycle information, including password age and accounts configured with non-expiring passwords.

```
File Edit View

Import-Module ActiveDirectory

Get-ADUser -Filter * -Properties DisplayName, SamAccountName, PasswordLastSet,
PasswordNeverExpires, Enabled |
Where-Object {$_.Enabled -eq $true} |
Select Name, SamAccountName, PasswordLastSet, PasswordNeverExpires |
Export-Csv C:\IAM-Audit-Lab\Reports\PasswordExpirationReport.csv -NoTypeInformation
```

```
Administrator: Windows Powe
PS C:\IAM-Audit-Lab\Scripts> .\PasswordExpirationReport.ps1
PS C:\IAM-Audit-Lab\Scripts> Import-Csv C:\IAM-Audit-Lab\Reports\PasswordExpirationReport.csv

Name                SamAccountName      PasswordLastSet      PasswordNeverExpires
-----
Administrator        Administrator        5/30/2026 3:38:54 PM True
Elie Nzweme           Elie Nzweme         5/30/2026 2:12:48 PM False
Bill Smith            bill.smith           6/10/2026 9:24:05 PM False
James Tritt           james.tritt          6/10/2026 9:26:13 PM False
John Williams         john.williams        6/10/2026 9:27:15 PM False
Janice Smith          janice.smith         6/10/2026 9:29:02 PM False
Jennifer Sorensen     jennifer.sorensen    6/10/2026 9:30:11 PM False
Julie Lloyd           julie.lloyd          6/10/2026 9:31:01 PM False
Katie Clark           katie.clark          6/10/2026 9:33:13 PM False
Oxana Wetherby        oxana.wetherby       6/10/2026 9:34:22 PM False
Henry Woodward        henry.woodward       6/10/2026 9:35:30 PM False
Kevin Jones           kevin.jones          6/10/2026 9:36:49 PM False
Lee McCrae            lee.mccrae           6/10/2026 9:37:48 PM False
Happy Sunday          happy.sunday         6/10/2026 9:38:42 PM False
Heather Summer        heather.summer       6/10/2026 9:40:07 PM False
Alyssa Jensen         alyssa.jensen        6/10/2026 9:41:42 PM False
Amy Wilcox            amy.wilcox           6/10/2026 9:42:39 PM False
Anthony Edwards       anthony.edwards      6/10/2026 9:43:39 PM False
Andrea Forsythe       andrea.forsythe      6/10/2026 9:44:38 PM False
Benjamin Washington   benjamin.washington  6/10/2026 9:45:41 PM False
Kathy Arnold          kathy.arnold         6/10/2026 9:47:11 PM False
Ted Williams          ted.williams          6/10/2026 9:48:03 PM False
William Trent         william.trent         6/10/2026 9:50:00 PM False
Shanon Tate           shanon.tate          6/11/2026 7:42:42 PM False
Trevor Anderson       trevor.anderson       6/11/2026 7:44:30 PM False
Trisha Hyatt          trisha.hyatt         6/11/2026 7:45:10 PM False
ITAdmin              itadmin              6/11/2026 8:04:58 PM False
svc_backup            svc_backup            6/11/2026 8:09:30 PM False
svcSplunk             svcSplunk            6/11/2026 8:10:19 PM False
svc_sql              svc_sql              6/11/2026 8:11:02 PM False
```

Output

PasswordExpirationReport.csv

PasswordExpirationReport - E...										
File Home Insert Page Layout Formulas Data Review View Help										
Clipboard Font Alignment Number Styles Cells Editing Add-ins										
A1 Name										
	A	B	C	D	E	F	G	H	I	J
1	Name	SamAccountName	PasswordLastSet	PasswordNeverExpires						
2	Administrator	Administrator	5/30/2026 15:38	TRUE						
3	Elie Nzweme	Elie Nzweme	5/30/2026 14:12	FALSE						
4	Bill Smith	bill.smith	6/10/2026 21:24	FALSE						
5	James Tritt	james.tritt	6/10/2026 21:26	FALSE						
6	John Williams	john.williams	6/10/2026 21:27	FALSE						
7	Janice Smith	janice.smith	6/10/2026 21:29	FALSE						
8	Jennifer Sorensen	jennifer.sorensen	6/10/2026 21:30	FALSE						
9	Julie Lloyd	julie.lloyd	6/10/2026 21:31	FALSE						
10	Katie Clark	katie.clark	6/10/2026 21:33	FALSE						
11	Oxana Wetherby	oxana.wetherby	6/10/2026 21:34	FALSE						
12	Henry Woodward	henry.woodward	6/10/2026 21:35	FALSE						
13	Kevin Jones	kevin.jones	6/10/2026 21:36	FALSE						
14	Lee McCrae	lee.mccrae	6/10/2026 21:37	FALSE						
15	Happy Sunday	happy.sunday	6/10/2026 21:38	FALSE						
16	Heather Summer	heather.summer	6/10/2026 21:40	FALSE						
17	Alyssa Jensen	alyssa.jensen	6/10/2026 21:41	FALSE						
18	Amy Wilcox	amy.wilcox	6/10/2026 21:42	FALSE						
19	Anthony Edwards	anthony.edwards	6/10/2026 21:43	FALSE						
20	Andrea Forsythe	andrea.forsythe	6/10/2026 21:44	FALSE						
21	Benjamin Washington	benjamin.washington	6/10/2026 21:45	FALSE						
22	Kathy Arnold	kathy.arnold	6/10/2026 21:47	FALSE						

7. Project Structure

Project Structure

Enterprise-IAM-Audit-Compliance-Lab

Enterprise-IAM-Audit-Compliance-Lab

Reports

DisabledUsers.csv
InactiveUsers.csv
LockedAccounts.csv
PasswordExpirationReport.csv
PrivilegedAccounts.csv
README.md

Screenshots

DisabledUsers.png
DisabledUserscommand.png
DisabledUserspowershelldisplay.png
InactiveUsers.png
LockedAccounts.png
PasswordExpirationReport.png
PrivilegedAccounts.png
Reports.png
adUsers.png
directories1.png
directories2.png
getmoduleavailablead.png
invalidcredentials.png
rsatadtools.png
rsatinstalled successfully.png
rsatinstalled successfully.png
scripts.png
verifyadmodule.png
verifydomainconnectivity.png

Scripts

DisabledUsers.ps1
InactiveUsers.ps1
LockedAccounts.ps1
PasswordExpirationReport.ps1
PrivilegedAccounts.ps1
README.md
LICENSE

Reports/

DisabledUsers.csv
InactiveUsers.csv
LockedAccounts.csv
PasswordExpirationReport.csv
PrivilegedAccounts.csv
README.md

List of disabled user accounts
List of inactive user accounts
List of locked user accounts
Password age and expiration report
Members of privileged groups
Project documentation

Screenshots/

DisabledUsers.png
DisabledUserscommand.png
DisabledUserspowershelldisplay.png
InactiveUsers.png
LockedAccounts.png
PasswordExpirationReport.png
PrivilegedAccounts.png
Reports.png
adUsers.png
directories1.png
directories2.png
getmoduleavailablead.png
invalidcredentials.png
rsatadtools.png
rsatinstalled successfully.png
scripts.png
verifyadmodule.png
verifydomainconnectivity.png

DisabledUsers report (Excel)
Command to generate disabled users
PowerShell output (disabled users)
Inactive users report (Excel)
Locked accounts report (Excel)
Password expiration report (Excel)
Privileged accounts report (Excel)
Reports folder screenshot
Active Directory Users and Computers
Directory structure screenshots
RSAT AD Tools check
Invalid credentials screenshot
RSAT AD Tools install command
RSAT AD Tools installed successfully
Scripts folder screenshot
Verify AD module
Verify domain connectivity

Scripts/

DisabledUsers.ps1
InactiveUsers.ps1
LockedAccounts.ps1
PasswordExpirationReport.ps1
PrivilegedAccounts.ps1
README.md
LICENSE

Script to find disabled users
Script to find inactive users
Script to find locked accounts
Script for password expiration report
Script to find privileged accounts
Project documentation
License information

Structure Overview

This project is organized into three main directories:

Reports

Contains CSV reports generated by the PowerShell audit scripts.

Screenshots

Contains evidence screenshots of commands, scripts, and report outputs.

Scripts

Contains all PowerShell scripts used to perform the IAM audits.

Legend

Folder

CSV Report (Excel)

PowerShell Script

Screenshot (Image)

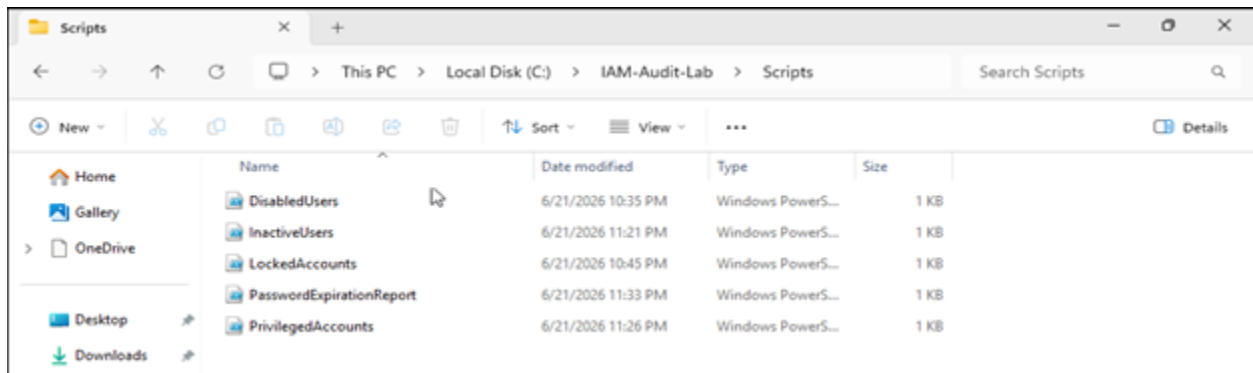
Documentation File

Purpose

This structure supports IAM audit activities, maintains documentation, and provides evidence for compliance and access governance reviews.

Scripts

Reports



8. Key IAM Controls Demonstrated

- User Lifecycle Management
- Account Governance
- Access Reviews
- Privileged Access Management (PAM)
- Account Lockout Monitoring
- Password Policy Auditing
- Active Directory Administration
- Security Reporting
- PowerShell Automation

9. Compliance Mapping

Framework		Control	Description
	NIST 800-53	AC-2	Account Management
	NIST 800-53	AC-6	Least Privilege
	CIS Controls v8	Control 5	Account Management
	ISO 27001	A.9	Access Control
	SOX	User Access Reviews	Privileged Account Monitoring

10. Skills Demonstrated

- Identity and Access Management (IAM)
- Active Directory Administration
- PowerShell Scripting
- Security Auditing
- Compliance Reporting
- Privileged Access Reviews
- User Account Management
- Access Governance

11. Audit Findings Summary

The IAM audit identified several account management and access governance findings within the Active Directory environment.

Audit Category



Key Findings

- Four disabled user accounts were identified and documented for review.
- Two user accounts were found in a locked-out state.
- Twenty-five user accounts showed no recent logon activity and were flagged as inactive.
- Three accounts were identified as members of the Domain Admins group and require periodic privileged access reviews.
- One account was configured with a non-expiring password and should be reviewed against organizational security policies.

Security Impact

Inactive, disabled, and privileged accounts present potential security risks if not reviewed regularly. Periodic IAM audits help ensure proper account lifecycle management, reduce unauthorized access risks, and support compliance requirements.

Recommended Actions

1. Review inactive accounts older than 90 days and disable or remove accounts no longer required.
2. Investigate account lockouts to determine whether they are caused by user error or suspicious activity.
3. Conduct quarterly privileged access reviews for all Domain Admin accounts.
4. Review accounts configured with non-expiring passwords and enforce password lifecycle policies where appropriate.
5. Maintain regular IAM audit reporting to support security and compliance objectives.

12. Project Metrics

Metric		Value
 Domain Controllers		1
 Client Systems		1
 Audit Scripts Developed		5
 CSV Reports Generated		5
 User Accounts Audited		34
 Disabled Accounts Found		4
 Locked Accounts Found		2
 Inactive Accounts Found		25
 Privileged Accounts Reviewed		3

13. Project Outcome

Successfully developed and executed automated IAM audit scripts that generated actionable security and compliance reports for Active Directory environments. The project demonstrates practical enterprise IAM auditing, PowerShell automation, and identity governance workflows commonly used by cybersecurity and infrastructure teams.

Source Code Repository:

<https://github.com/elienczweme/Enterprise-IAM-Audit-Compliance-Lab>